



Finding missing angles

IF THE LENGTHS OF TWO SIDES OF A RIGHT TRIANGLE ARE KNOWN, ITS MISSING ANGLES CAN BE FOUND.

To find the missing angles in a right triangle, the inverse sine, cosine, and tangent are used. Use a calculator to find these values.

Which formula?

Choose the formula that contains the pair of sides that are given in an example. For instance, use the sine formula if the lengths of the hypotenuse and the side opposite the unknown angle are known, and the cosine formula if the lengths of the hypotenuse and the side next to the angle are given.

$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$$

△ The sine formula

Use the sine formula if the lengths of the hypotenuse and the side opposite the missing angle are known.

$$\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$$

△ The cosine formula

Use the cosine formula if the lengths of the hypotenuse and the side adjacent (next to) to the missing angle are known.

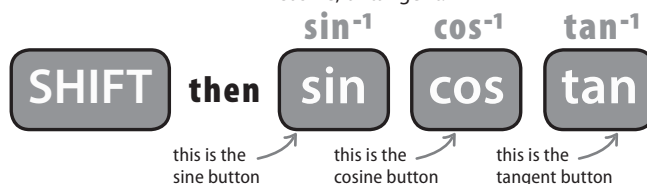
$$\tan A = \frac{\text{opposite}}{\text{adjacent}}$$

△ The tangent formula

Use the tangent formula if the lengths of the sides opposite and adjacent to the missing angle are known.

▽ Calculator functions

To find the inverse values of sine, cosine, and tangent, press shift before sine, cosine, or tangent.



SEE ALSO

◀ 72–73 Using a calculator

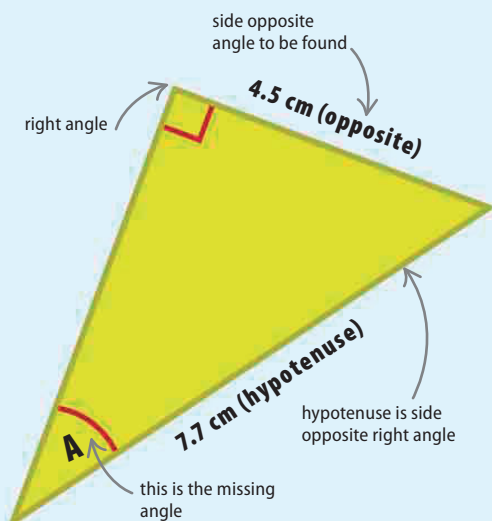
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Using the sine formula

In this right triangle the hypotenuse and the side opposite angle A are known. Use the sine formula to find the size of angle A.



Choose the right formula—in this example the hypotenuse and the side opposite the missing angle, A, are known, so use the sine formula.

Substitute the known values into the sine formula.

Work out the value of sin A by dividing the opposite side by the hypotenuse.

Find the value of the angle by using the inverse sine function on a calculator.

Round the answer to a suitable size. This is the value of the missing angle.

$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin A = \frac{4.5}{7.7}$$

answer is rounded to 4 decimal places

$$\sin A = 0.5844$$

press shift then the sine button to get inverse sine

$$A = \sin^{-1}(0.5844)$$

$$A = 35.76^\circ$$

this is rounded to 2 decimal places